A Comparative Evaluation of Encoder and Decoder Language Models with Parameter-Efficient Fine-Tuning (PEFT/LoRA) for Multi-Class Text Emotion Recognition

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***Abstract*—Text-based emotion recognition remains challenging in Natural Language Processing due to severe class imbalance, contextual ambiguity, and informal language in short digital texts. Prior work has compared individual encoder or decoder architectures in isolation, but it remains unclear which factor most shapes the trade-off between classification quality and computational cost: the underlying architectural family (recurrent, encoder-Transformer, or decoder-Transformer), the parameter scale within a family, or the training objective used during pre-training. This work proposes a controlled, three-axis experimental framework to address this gap: a non-neural and a recurrent baseline, fully fine-tuned encoder Transformers (DistilBERT, BERT-Base), and decoder Transformers (Qwen2.5-0.5B/1.5B, Gemma 4 E2B/E4B, DeepSeek-R1-Distill-Qwen-1.5B) adapted via Parameter-Efficient Fine-Tuning (PEFT/LoRA), evaluated on the CARER emotion dataset under matched hardware constraints. Rather than identifying a single best-performing model, our objective is to characterize which of these three factors, architectural family, parameter scale, or training objective, most strongly shifts the Macro-F1-versus-compute trade-off in multi-class emotion classification. We detail this experimental design and its grounding in the current literature, providing a methodological basis for practitioners choosing between compact encoders and LoRA-adapted decoders under resource constraints.**

***Index Terms*—Text Emotion Recognition, Large Language Models, LoRA, PEFT, Macro-F1, Multi-class Classification.**

I. Introduction

Text-based emotion recognition is central to applications ranging from conversational agents to consumer feedback analysis and mental health monitoring. However, reliably detecting fine-grained emotional states in brief, informal digital texts remains difficult due to contextual ambiguity and severe class imbalance. While recurrent models like BiLSTM and encoder-based architectures like BERT have long served as primary baselines, generative Large Language Models (LLMs) have demonstrated strong potential in capturing complex semantic variations through advanced pre-training. Despite this potential, the massive parameter size of modern decoders makes full fine-tuning computationally unfeasible for resource-constrained environments, highlighting the need for efficient adaptation techniques.

Parameter-Efficient Fine-Tuning (PEFT) methods, particularly Low-Rank Adaptation (LoRA), address these hardware bottlenecks by freezing the pre-trained base model and updating only a small subset of low-rank adapter weights. Although decoders such as Qwen, Gemma, and DeepSeek excel in general language tasks, their performance when adapted via LoRA for specialized downstream classification under high class imbalance is not yet fully understood. Existing benchmarks often focus on broad sentiment analysis or general classification, relying on overall accuracy, a metric that frequently masks poor performance on minority emotional classes. Furthermore, direct comparisons between fully fine-tuned lightweight encoders and LoRA-adapted intermediate decoders under identical hardware constraints remain scarce in current literature.

To address these gaps, this work evaluates the trade-offs between legacy encoder baselines and modern intermediate-scale decoders on the highly imbalanced CARER emotion dataset.

Building on this motivation, our guiding research question is the following: how do architectural family, parameter scale, and training objective each affect the trade-off between classification quality (Macro-F1) and computational cost in multi-class emotion classification, and which of these three factors most strongly influences that trade-off? Rather than searching for a single winning architecture, this work isolates each factor through matched comparisons, allowing its individual contribution to be characterized independently of the others.

Our main contributions are as follows:

- **Three-Axis Controlled Benchmark:** We design a controlled comparison that isolates architectural family, parameter scale, and training objective as independent variables, rather than benchmarking architectures in an unstructured manner.

- **Efficiency & Optimization Analysis:** We evaluate the operational trade-offs of PEFT/LoRA across decoder architectures, focusing on memory footprint, parameter count, and convergence rates under strict hardware limitations.
- **Imbalance-Aware Evaluation:** We analyze model resilience to severe class imbalance on the CARER dataset, prioritizing Macro-F1 metrics to ensure fair assessment across underrepresented emotion classes.
- **Practical Deployment Guidelines:** We outline concrete performance-versus-compute trade-offs to offer actionable decision frameworks for researchers and practitioners choosing between compact encoders and LoRA-adapted decoders.

The remainder of this survey is structured as follows: Section II reviews the evolution and current approaches in text-based emotion recognition. Section III analyzes parameter-efficient adaptation methods across Encoder and Decoder architectures. Section IV examines the benchmark dataset characteristics and evaluation protocols. Section V outlines our proposed experimental setup and methodology for future empirical training. Section VI discusses preliminary trade-offs and future research directions, and Section VII concludes the paper.

II. Domain Analysis & Evolution of Text Emotion Recognition

A. Dominant Traditional Approaches

Text-based emotion recognition (TBED) has evolved through a rapid progression of computational methodologies designed to address semantic ambiguity and unstructured language. Affective computing has transitioned from rule-based and lexicon-driven heuristics toward deep representation learning and generative paradigms [1]. These foundational techniques can be broadly categorized into keyword-based, corpus-based, rule-based, traditional machine learning, and deep learning: Some of these computational approaches have relied on multiple computational paradigms to interpret a text and determine which emotions are shown. These foundational techniques can be broadly categorized into keyword-based, corpus-based, ruled-based, tradicional machine learning and deep learning.

- **Keyword-Based Approaches**: Identify explicit emotional terms using lexical databases (such as WordNet-Affect or SentiWordNet) and apply linguistic rules to assess sentiment intensity [2].
- **Corpus-Based Approaches**: Induce domain-specific word-emotion lexicons using statistical methods and generative models across annotated or weakly-labeled text datasets [2].
- **Rule-Based Approaches**: Apply expert heuristic rules, Part-of-Speech tagging, and phrasal patterns to infer emotional states based on predefined linguistic structure [2].
- **Traditional Machine Learning Approaches**: Utilize supervised algorithms, such as, Support Vector Machines (SVM), Naïve Bayes, Decision Trees, and K-Nearest Neighbors that rely on manual feature engineering to classify text into emotional categories [2].
- **Deep Learning Approaches**: Employ neural network architectures, such as, Recurrent Neural Networks (RNNs/LSTMs), Convolutional Neural Networks (CNNs), and Transformer-based models (e.g., BERT), to automatically learn hierarchical representations and capture complex contextual semantics directly from raw text [1].

B. The Shift to Generative Decoders

The rise of generative Large Language Models (LLMs) has fundamentally changed how we approach text classification in NLP [3]. Traditionally, affective computing tasks relied on discriminative models like BERT or GRU variants that learn a direct mapping function $f(X) \to Y$ from input text X to a predefined set of labels Y [4]. In contrast, autoregressive decoder architectures treat classification as a conditional text generation problem. They model the probability of generating a target class token y given a prompt sequence X:

$$P(y \mid X) = \prod_{i=1}^{N} P(w_i \mid w_1, \dots, w_{i-1}, X) \tag{1}$$

This shift offers significant benefits, especially when it comes to leveraging broad contextual knowledge and picking up on subtle emotional tone that standard encoder representations often miss [5].

1) Zero-Shot and Few-Shot Prompting Paradigms: A key feature of generative decoders is their ability to handle downstream classification without updating model weights, relying instead on in-context learning (ICL) [6]. In a zero-shot setup, the model gets an instruction template describing the task along with the input text, using its pre-trained knowledge to output the predicted label. Few-shot prompting builds on this by adding k example pairs directly into the prompt context:

$$\text{Prompt} = \{(X_1, Y_1), (X_2, Y_2), \dots, (X_k, Y_k), X_{\text{test}}\} \tag{2}$$

While zero-shot and few-shot setups avoid task-specific training costs, their reliability for fine-grained emotion tasks varies greatly. Zero-shot decoders are notoriously sensitive to how prompts are written, how context examples are ordered, and how subtle affective context is presented. Furthermore, without parameter adaptation for a specific domain, zero-shot LLMs often struggle with subtle emotion boundaries. Furthermore, empirical studies comparing small task-specific encoders against 7B generative models (e.g., Llama-2 and Mistral) show that in-context prompting often yields higher variance and lower overall classification accuracy on affective tasks compared to direct supervised tuning [7].

2) Supervised Fine-Tuning versus In-Context Learning: To address the instability of in-context learning, supervised fine-tuning (SFT) adjusts the model's output probabilities to align directly with task-specific labels. In multi-class emotion recognition, SFT restricts the output space to valid emotion categories while teaching the model to handle informal digital slang and implicit emotional cues [8].

However, choosing between prompting and full fine-tuning presents clear trade-offs:

- **In-Context Learning (Zero/Few-Shot):** Keeps base weights frozen and avoids training costs, but can struggle with subtle emotional nuances.
- **Full Supervised Fine-Tuning (SFT):** Delivers strong task alignment across imbalanced emotion classes, but requires updating billions of parameters ($> 10^9$), creating massive computational and memory bottlenecks during backpropagation.

This balance demonstrates why intermediate approaches are necessary. Generative decoders provide strong semantic understanding, but using them effectively for emotion classification requires parameter-efficient strategies, such as Low-Rank Adaptation (LoRA), that deliver SFT performance without excessive computational demands [9].

C. Established Standards and Open Challenges

Sentiment analysis and emotion recognition overlap significantly in computational linguistics, but they present very different levels of complexity. Distinguishing basic sentiment classification from fine-grained emotion recognition makes it easier to pinpoint the key challenges facing modern NLP.

1) Established Baselines in Binary Sentiment Analysis: Binary sentiment classification, which sorts text into simple positive or negative categories, is a well-established task. Earlier studies applying fine-tuned Encoders like BERT alongside GRU variants reached near-peak performance on benchmarks such as SST-2 and IMDb [4]. Because these datasets rely on balanced classes and explicit sentiment words, standard supervised models handle well-structured text with high reliability.

2) Open Challenge 1: Fine-Grained Multi-Class Emotion Mapping: Moving beyond binary sentiment requires mapping text to specific psychological states, such as Ekman's six core emotions: anger, fear, joy, love, sadness, and surprise. As categorized in emotion detection surveys [1], adopting a discrete theoretical model simplifies categorical boundaries but introduces challenges when modeling overlapping affective states. Distinct emotions like fear and surprise frequently share identical keywords, meaning the model must rely on broader context rather than individual terms to tell them apart [8]. Modern decoder architectures capture these subtle contextual shifts well, but mapping their outputs into discrete emotion categories without misclassifications remains difficult [6].

3) Open Challenge 2: Extreme Class Imbalance in Affective Corpora: A major bottleneck in emotion datasets like the CARER benchmark is severe class imbalance [5]. Everyday language features common emotions like joy or sadness far more often than high-intensity states like surprise or fear. Standard cross-entropy loss naturally focuses on majority-class errors, which leads to high overall accuracy but poor recall on minority categories [1]. While sampling methods and weighted losses help traditional encoders, managing severe imbalance within Parameter-Efficient Fine-Tuning (PEFT) frameworks, where base weights remain frozen, is still an ongoing challenge for Macro-F1 optimization [9].

4) Open Challenge 3: Ambiguity in Digital Text: Short online texts introduce noise that degrades classification accuracy. Much of this noise is produced by the way people write, there are writing styles that can lead to misinterpretations in literary readings [1]. The main causes of ambiguity in affective corpora include:

- **Implicit Emotions and Sarcasm:** Phrases without direct emotion keywords (such as *"Great, another delay."*) depend on pragmatic reasoning that basic encoders often miss.
- **Informal Language:** Heavy use of slang, shortcuts, irregular grammar, and repeated punctuation (like *"???"*) creates out-of-vocabulary problems during tokenization.
- **Limited Context:** Brief posts provide very little surrounding text, making it harder to determine the exact emotion of words that change meaning based on subtext.

Balancing small, efficient architectures against the deep contextual understanding needed to resolve these ambiguities within strict memory limits remains a major objective for practical emotion recognition systems.

III. Parameter-Efficient Adaptation of Encoder and Decoder Architectures

A. Full Fine-Tuning vs. PEFT Paradigms

Supervised Fine-Tuning (SFT) has historically served as the standard paradigm for adapting pre-trained language models to downstream tasks. In full parameter fine-tuning (FFT), every weight matrix Θ across all model layers is updated via backpropagation. For compact encoder architectures such as DistilBERT ($\approx$ 66M parameters) and BERT-Base ($\approx$ 110M parameters), full parameter updates are computationally lightweight, requiring under 2 GB of VRAM during training [4]. Consequently, FFT remains a highly effective baseline for standard encoder-based classification tasks.

However, as model scales transition toward intermediate and large-scale generative decoders (e.g., Qwen-2.5-1.5B/7B, Gemma-2-2B/9B, and DeepSeek-R1-Distill-1.5B/7B), full parameter updates become computationally prohibitive [10]. In 16-bit floating-point precision (FP16/BF16), storing a 7-billion parameter model consumes approximately 14 GB of static memory. During full fine-tuning, the memory footprint scales drastically when factoring in first- and second-order optimizer states (such as AdamW tracking momentum and variance for every weight), gradient buffers, and activation caching:

$$\text{MemoryFFT} \approx \Theta\text{weights} + \Theta_{\text{gradients}} + \Theta_{\text{optimizer states}} + \text{Activations} \tag{3}$$

This memory requirement easily exceeds 80 GB of peak VRAM for a 7B model, far surpassing the memory limits of standard consumer GPUs or single-workstation academic environments [9].

To mitigate these memory bottlenecks, Parameter-Efficient Fine-Tuning (PEFT) freezes the pre-trained base parameters

Θ_0 and restricts gradient updates to a small set of auxiliary parameters Φ, where $|\Phi| \ll |\Theta_0|$ (typically $< 1.5\%$ of the base parameter count) [9], [11].

1) Taxonomy of Parameter-Efficient Adaptation: PEFT strategies can be broadly categorized into three core architectural paradigms based on how auxiliary parameters are integrated into the network [10]:

- **Addition-Based Methods:** Introduce lightweight, trainable structural modules (such as sequential or parallel adapter layers) or continuous soft prompt tokens directly into the model architecture while keeping base weights entirely frozen [10].
- **Selection-Based Methods:** Freeze the vast majority of original weights and selectively unfreeze a small, highly targeted sub-network or specific parameter subsets (e.g., bias vectors in BitFit) for gradient updates [10].
- **Reparameterization-Based Methods:** Exploit the intrinsic low-rank properties of weight updates during adaptation. By parameterizing weight shifts ΔW through low-rank matrix factorizations ($W = W_0 + BA$), methods like Low-Rank Adaptation (LoRA) enable efficient adaptation without altering base model inference latency [9], [10].

2) Empirical Performance Bounds and Robustness Trade-offs: While PEFT drastically reduces hardware overhead, comparative literature highlights trade-offs regarding raw task capability, instruction-following proficiency, and adversarial resilience when compared to full fine-tuning [10].

On multi-turn instruction benchmarks, empirical evaluations demonstrate that full parameter tuning maintains a performance margin over parameter-efficient alternatives across various architectural scales [10]:

- **TinyLLaMA Scale:** Full fine-tuning achieves an average MT-Bench score of 2.68, outperforming LoRA (2.05) and Prefix-Tuning (1.81) [10].
- **Mistral-7B Scale:** FFT reaches an average score of 4.96, maintaining an advantage over LoRA (4.74) and Prefix-Tuning (4.67) [10].
- **LLaMA2-7B Scale:** FFT achieves an average MT-Bench score of 5.26, compared to 4.91 for LoRA and 4.82 for Prefix-Tuning [10].

Furthermore, evaluations on the AdvGLUE benchmark indicate that while PEFT methods perform comparably to FFT on clean validation sets, full fine-tuning demonstrates superior structural robustness under adversarial perturbations [10]:

- **RoBERTa-Base (Adversarial Noise):** Under adversarial attacks, FFT retains an average robustness score of 36.58, significantly outperforming LoRA (30.89) and BitFit (28.21) [10].
- **RoBERTa-Large (Adversarial Noise):** FFT achieves an overall robustness score of 55.44, compared to LoRA's 53.96 [10].

3) Scope Justification for Text Emotion Recognition: Despite the empirical edge held by FFT in highly complex or adversarial environments, PEFT represents an optimal adaptation framework for multi-class text emotion recognition. Fine-grained emotion classification on social media corpora relies primarily on capturing localized semantic cues, pragmatic context, and affective syntax rather than performing unconstrained open-ended text generation or complex multi-step reasoning. Consequently, the reduced parameter subspace updated via LoRA ($r = 16, \alpha = 32$) provides sufficient capacity to resolve affective ambiguities while allowing intermediate decoders to operate efficiently within constrained compute environments [9]

B. Mechanics of Low-Rank Adaptation (LoRA)

Low-Rank Adaptation formalizes the reparameterization strategy introduced in Section III.A by constraining the weight update ΔW of a pre-trained matrix $W_0 \in \mathbb{R}^{d\times k}$ to a low-rank decomposition. Instead of learning a dense update matrix with $d \times k$ free parameters, LoRA expresses ΔW as the product of two much smaller matrices:

$$\Delta W = BA, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},\ r \ll \min(d,k) \tag{4}$$

During the forward pass, the frozen base projection and the trainable low-rank branch are combined additively, scaled by a constant factor:

$$h = W_0 x + \frac{\alpha}{r} BAx \tag{5}$$

where α is a fixed scaling hyperparameter. Normalizing by r keeps the effective magnitude of the update stable when the rank is changed, avoiding the need to re-tune the learning rate every time r is adjusted [9].

1) Initialization Scheme: To guarantee that adaptation begins from the exact pre-trained behavior of the base model, LoRA follows an asymmetric initialization: matrix A is initialized from a random Gaussian distribution, while matrix B is initialized to zero. Consequently, $\Delta W = BA = 0$ at the start of training, and the model's output is identical to the unadapted base model before any gradient step is taken.

2) Target Module Selection: LoRA adapters are not injected into every weight matrix of a Transformer block; instead, they are selectively applied to the linear projections most responsible for task-specific behavior. As detailed later in Table VI, this work targets the attention projections W_q, W_k, W_v, W_o across all decoder architectures, extending to the feed-forward projections ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$) for DeepSeek-R1-Distill-Qwen-1.5B specifically. This selective targeting keeps the trainable parameter count within 1–1.5% of the base decoder size [9], [11].

3) Why a Low-Rank Update Suffices: The effectiveness of this constrained parameterization rests on the intrinsic-rank hypothesis: adapting a large pre-trained model to a related downstream task does not require exploring its full parameter space, but only a low-dimensional subspace of it [10]. LoRA operationalizes this hypothesis directly, rather than leaving it as an implicit property of the optimization trajectory, which is precisely the restricted solution space discussed in Section III.A. This restricted-subspace view aligns with broader evidence that pre-trained language models develop identifiable internal structures during training, structures that downstream adaptation methods can exploit rather than relearn from scratch [12]. Efficiency-oriented alternatives that depart from the Transformer architecture altogether, such as spiking neural networks distilled from BERT, pursue a related goal through a different mechanism, replacing rank-constrained weight updates with sparse, event-driven computation; such architectures nonetheless fall outside the scope of the LoRA-based adaptation examined in this work [13].

C. Comparative Methodologies: LoRA vs. QLoRA vs. Full Fine-Tuning

Building on the mechanics introduced in Section III.B, this section positions Low-Rank Adaptation and its quantized variant, QLoRA, relative to full fine-tuning along a single axis: how much memory a given adaptation strategy requires to reach comparable task performance.

1) Quantization-Aware Adaptation: QLoRA: QLoRA extends the reparameterization scheme described above by additionally storing the frozen base weights W_0 in a 4-bit quantized format (commonly NormalFloat4, NF4), rather than the standard 16-bit (FP16/BF16) precision used by conventional LoRA. Because W_0 remains frozen throughout training, this quantization affects only the static memory footprint of the base model, not the trainable low-rank matrices A and B, which are still updated in higher precision. As summarized in Table VI, this work applies 4-bit NormalFloat quantization specifically on the lower-VRAM GPUs available to the team, reducing the static weight memory of the decoder models by approximately fourfold relative to FP16 storage [14], at the cost of a small computational overhead introduced by on-the-fly dequantization during the forward and backward passes.

2) Positioning Along the Memory-Performance Spectrum: The three adaptation strategies considered in this work occupy distinct points along a single memory-performance spectrum:

- **Full Fine-Tuning (FFT):** Updates every parameter in Θ, requiring the largest memory footprint of the three but retaining the highest empirical ceiling on task performance and adversarial robustness [10].
- **LoRA:** Freezes Θ_0 and restricts updates to the low-rank matrices A, B, reducing trainable parameters to under 1.5% of the base model while storing the frozen weights at standard precision [9].
- **QLoRA:** Combines the LoRA reparameterization with 4-bit quantization of Θ_0, further reducing static memory at the cost of a modest computational overhead, without changing the number of trainable parameters [14].

3) Architectural Compatibility Across Model Families: Because LoRA and QLoRA target the same class of linear projection matrices described in Section III.B, both methods apply uniformly across the encoder and decoder architectures evaluated in this work. The attention projections W_q, W_k, W_v, W_o are present in BERT-based encoders as well as in the Qwen, Gemma, and DeepSeek decoders, so the adaptation mechanism itself does not need to change across families. What differs in practice is how much quantization matters: for the encoder baselines (DistilBERT, BERT-Base), which are fully fine-tuned under 2 GB of VRAM [4], quantization offers little practical benefit, while for the decoder models, whose base weights alone approach or exceed the available VRAM on the team's lower-end GPUs, QLoRA is what makes training feasible at all [14].

D. Theoretical Limitations and Underlying Assumptions

1) The Intrinsic-Rank Assumption Is Empirical, Not Guaranteed: The intrinsic-rank hypothesis introduced in Section III.B is supported by empirical observation, not by a closed-form guarantee that the optimal weight update for a given task actually lies within a rank-r subspace. The rank $r = 16$ used throughout this work is a fixed hyperparameter chosen a priori, not derived from a task-specific theoretical bound; if the true update required for multi-class emotion classification exceeds this rank, LoRA would underfit silently, and the resulting performance gap could be misattributed to architectural family or training objective rather than to a capacity mismatch introduced by the adaptation method itself. This variance across target modules and tasks has already been observed empirically [9], but its theoretical origin remains only partially understood.

2) Generalization Across Architectures and Objectives Is Not Assumed: The theoretical grounding for low-rank adaptation, including the view that pre-trained models develop structures a downstream update can exploit rather than relearn [12], was developed primarily on general-purpose language models rather than validated separately across recurrent, encoder, and decoder families pretrained under distinct objectives. Because this work's central question is precisely whether architectural family, parameter scale, or training objective most shapes the quality-cost trade-off, assuming in advance that LoRA behaves identically across all three would presuppose the answer this study sets out to test. This assumption is therefore treated here as a hypothesis under evaluation, not as an established premise.

3) Robustness Bounds Derived from General NLP Tasks: The performance and robustness bounds separating full fine-tuning from PEFT methods discussed in Section III.A, including the adversarial robustness gap reported for RoBERTa-Large (55.44 vs. 53.96) [10], were measured on structured reasoning and generation benchmarks such as GSM8k and SQL, not on short, informal, class-imbalanced affective text. Whether the same gap holds for multi-class emotion classification on the CARER benchmark is an open empirical question that this study's experimental design directly addresses, rather than one it assumes settled in advance.

4) Instability and Quantization as Compounding Sources of Error: Adversarial training results show that the sparse trainable-parameter regime of standard LoRA introduces measurable instability under distribution shift and low-resource conditions [15], while separate survey evidence flags a tendency toward overconfident predictions when PEFT-tuned models are applied to small or imbalanced datasets [14]. Because QLoRA's 4-bit quantization of the frozen base weights is applied on top of an already parameter-scarce adapter, this work implicitly assumes that rank restriction and numerical quantization introduce independent, additive sources of approximation error rather than interacting nonlinearly; this interaction has not been directly characterized in the reviewed literature and is acknowledged here as an open limitation.

E. Cross-Dataset Perspective: EMO-KNOW

To understand how emotion recognition models perform on larger and more detailed tasks, it is helpful to look at other benchmark datasets. A key example is EMO-KNOW, which consists of roughly 700,000 tweets labeled across 48 emotion categories [16]. Beyond basic categories like joy, sadness, and anger, EMO-KNOW includes more specific affective states such as hope, despair, and relief [16]. This dataset serves as a useful benchmark for evaluating fine-grained emotion classification.

In their evaluation of EMO-KNOW, Jayanthi and Kavitha [16] compared several machine learning and deep learning models. Logistic Regression achieved a solid 92.0% accuracy, while Transformer models like BERT and RoBERTa reached 93.0% and 94.0%, respectively [16]. The highest performance came from a hybrid TF-IDF + LSTM model, which achieved 95.4% accuracy [16]. These results show that combining traditional frequency-based features with contextual sequence modeling is an effective strategy for complex emotion detection.

The authors also emphasized the impact of text preprocessing when dealing with social media content. Standard cleaning steps, such as removing punctuation, hashtags, user mentions, and stopwords, as well as applying stemming or lemmatization, which led to noticeable accuracy improvements across models [16]. This step is particularly important for short social media posts, where informal phrasing and limited context add to the classification difficulty.

Although EMO-KNOW and CARER differ in total size and label count (48 classes versus 6), both datasets reflect the core challenge of identifying emotions in informal online text. Looking at the results from EMO-KNOW provides a helpful external reference when examining how different model families behave. However, direct accuracy comparisons should be avoided, as the datasets, class distributions, and experimental setups are not the same.

F. Methodological and Architectural Alternatives in Literature

To address feature representation limitations and improve structural adaptability, several hybrid and multi-layer strategies have been established in affective computing literature [17]:

a) Hybrid Feature Architectures:

- **1D CNN-BiLSTM Classification Heads:** Instead of relying solely on standard linear projections, integrating a parallel 1D CNN-BiLSTM network on top of transformer embeddings enables the concurrent extraction of localized n-gram emotional triggers (via CNN) and long-range sequential dependencies (via BiLSTM) [17].
- **Multi-Layer Representation Aggregation:** Rather than using exclusively the final hidden state or pooled `[CLS]` token, feature representations can be constructed by concatenating activations across the final four transformer layers, extracting richer hierarchical semantics prior to task projection [17].

b) Granular Error and Multi-Corpus Diagnostics:

- **Semantic Pair Confusion Analysis:** Expanding performance evaluations to analyze misclassification dynamics between semantically adjacent emotion categories (e.g., *fear* vs. *surprise*, *joy* vs. *love*) isolates structural failure modes in short-text affective processing [17].
- **Cross-Domain Benchmarking:** Evaluating models across multi-party conversational corpora with varied dialogue lengths validates whether performance and computational efficiency trade-offs remain consistent beyond isolated, single-sentence benchmarks.

IV. Benchmark Analysis: The CARER Dataset, & Evaluation Protocols

A. Corpus Characteristics and Domain Scope

To evaluate multi-class text emotion recognition under realistic conditions, we utilize the CARER benchmark dataset (hosted as the "Emotions: Where Words Paint the Colors of Feelings" corpus on Kaggle). The corpus consists of short, self-reported English social media posts and personal narratives annotated for affective state.

1) Class Distribution and Imbalance: The dataset comprises $20,000$ short text instances categorized into six distinct emotional classes based on psychological models of primary emotion: *joy*, *sadness*, *anger*, *fear*, *love*, and *surprise*. As illustrated in Table I, the dataset exhibits severe class imbalance.

TABLE I
CLASS DISTRIBUTION ACROSS THE EVALUATION CORPUS

Emotion Class	**Sample Count (N)**	**Percentage (%)**
Joy	$6,761$	33.80%
Sadness	$5,797$	28.99%
Anger	$2,159$	10.80%
Fear	$1,937$	9.69%
Love	$1,641$	8.20%
Surprise	$1,705$	8.52%
Total	$20,000$	100.00%

Majority classes (*joy* and *sadness*) account for over 62% of the total corpus, while minority affective states such as *love* and *surprise* collectively represent less than 17%. This structural skew makes the dataset an ideal testbed for evaluating model resilience to class imbalance and justifies prioritizing unweighted Macro-F1 metrics over raw accuracy.

2) Linguistic Complexity and Domain Scope: The text entries exhibit domain-specific characteristics typical of informal digital communication:

- **Short Sequence Lengths:** Posts average between 15 and 30 words, providing limited local context for disambiguating emotional states.

- **Informal Syntax and Noise:** Heavy reliance on colloquial language, irregular punctuation, implicit emotional expressions, and social media slang.
- **Overlapping Class Boundaries:** Fine-grained distinctions between closely related affective categories (e.g., distinguishing high-arousal *fear* from *surprise*, or *joy* from *love*) present high contextual ambiguity.

These properties establish a challenging benchmark where models must capture subtle semantic nuances while maintaining balanced classification performance across underrepresented emotion classes.

B. Reported Metrics and Published Baselines

Evaluating fine-grained emotion recognition models on imbalanced corpora requires selecting evaluation metrics that accurately reflect performance across all target categories while accounting for resource consumption [4], [5].

1) Evaluation Metrics: To measure both classification performance and operational cost, models are evaluated across a standardized set of metrics:

- **Macro-F1 ($\mathbf{F_1^{macro}}$):** Serves as the primary quality metric. By calculating the unweighted mean of F_1 scores across all six emotion categories, Macro-F1 treats every class equally, making it sensitive to poor performance on underrepresented classes like *surprise* or *love* [1], [5].
- **Weighted-F1 ($\mathbf{F_1^{weighted}}$):** Computes the average F_1 score weighted by class support. While provided for comparison, it can mask weak performance on minority categories when majority classes achieve high precision and recall [3].
- **Inference Latency and Memory Footprint:** Measures average prediction time per sample (in milliseconds) alongside peak GPU memory utilization (in gigabytes) to quantify operational deployment overhead [9].

2) Published Baselines and Benchmark References: To establish performance bounds, our results are benchmarked against traditional baselines and previously published transformer results on the CARER dataset structure [4], [5]:

TABLE II
REPRESENTATIVE PERFORMANCE BASELINES ON CARER

Model	**Macro-F1**	**Weighted-F1**	**Characteristics**
TF-IDF + LogReg	$\sim 78.5\%$	$\sim 82.1\%$	Fast CPU baseline; no context
BiLSTM	$\sim 84.2\%$	$\sim 86.0\%$	Recurrent baseline [5]
DistilBERT	$\sim 89.8\%$	$\sim 91.2\%$	Efficient encoder [4]
BERT-base	$\sim 91.5\%$	$\sim 92.6\%$	Strong encoder baseline [4]

Traditional non-neural and recurrent baselines establish the lower performance boundary [5], [8]. While lightweight and fast to train, they struggle to capture the complex pragmatic cues and implicit emotional expressions present in digital text [3]. Encoder-based architectures like BERT-Base set a strong performance baseline for discriminative models [4].

Our experimental framework builds directly upon these references, testing whether parameter-efficient generative decoders (such as Qwen2.5, Gemma-2, and DeepSeek-R1-Distill) can match or surpass these established encoder boundaries while operating under strict hardware and parameter budgets [9], [11].

C. Known Dataset Limitations and Biases

While the CARER benchmark provides a structured framework for evaluating multi-class emotion recognition [5], several intrinsic dataset limitations and distributional biases must be acknowledged to contextualize model performance.

1) Linguistic and Register Constraints: The dataset consists exclusively of English-language text compiled from social media posts and self-reported affective narratives [5]. This introduces specific domain boundaries:

- **Monolingual Scope:** Models evaluated on this corpus are restricted to English input, limiting the generalizability of downstream findings to multilingual or cross-lingual affective tasks.
- **Informal Register Bias:** Text instances predominantly exhibit informal syntax, colloquialisms, non-standard punctuation, and social media slang. Consequently, representations may show degraded robustness when transferred to formal prose, clinical transcripts, or technical domains.

2) Annotation Granularity and Cultural Bias: The corpus adopts a discrete taxonomy of six primary emotion categories based on psychological models of affect [5]. This introduces structural simplifications:

- **Discrete Categorization of Complex States:** Real-world human emotional expression frequently involves blended or context-dependent emotional states (e.g., bittersweetness, anxious surprise) that cannot be fully captured by mutually exclusive single-label assignments [6].
- **Cultural Expression Bias:** Because annotations align primarily with Western psychological constructs and English-speaking demographic norms, subtle cultural variations in implicit emotional framing may introduce annotation noise or model classification bias [3].

Addressing these domain-specific constraints requires careful empirical error analysis during validation, ensuring that strong benchmark performance on Macro-F1 is not mistaken for universal affective generalization across broader linguistic registers.

V. PROPOSED EXPERIMENTAL METHODOLOGY

A. Experimental Framework and Hardware Constraints

To rigorously assess the trade-offs between classification quality and computational cost in multi-class text emotion recognition, we establish a standardized experimental environment. The evaluation spans a representative spectrum of model families, parameter scales, and pre-training objectives across both non-neural and neural paradigms.

TABLE III
COMPARATIVE SUMMARY OF EXISTING APPROACHES IN AFFECTIVE COMPUTING AND PEFT ARCHITECTURES

Reference	Year	Technique	Dataset / Environment	Primary Metric	Reported Result	Identified Limitation
Saravia et al. [5]	2018	BiLSTM + Contextual Word Embeddings	CARER (Emotion Benchmark)	Macro-F1	0.647 (Benchmark Baseline)	Limited long-range semantic capacity; struggles with severe class imbalance.
Kratzwald et al. [8]	2018	Deep Learning (BiLSTM / CNN)	Financial & Customer Reviews	Classification Accuracy	82.4% Accuracy	High dependence on task-specific fine-tuning; poor zero-shot generalization.
Chakriswaran et al. [3]	2019	Lexicon-Based & Deep Learning Survey	Multimodal / Microblogs	Accuracy / F1-Score	Qualitative SOTA Overview	Highlights systemic issues with informal slang, sarcasm, and class skew.
Zhang [4]	2024	BERT, RNN, GRU Full Fine-Tuning	Standard Sentiment Corpora	Accuracy / Macro-F1	BERT outperforms RNN/GRU ($> 90\%$)	Full fine-tuning requires substantial memory; limited to fixed context lengths.
Nwaiwu [9]	2025	LoRA, $(IA)^3$, ReFT Evaluation	Low-Resource Text Benchmarks	Macro-F1 / Memory Footprint	LoRA achieves $\sim 98\%$ of full fine-tuning	Performance varies significantly across adapter target modules and target tasks.
Mohammed & Kora [11]	2026	LLMs + LoRA (PEFT)	Text-Based Deepfake Corpora	Detection Accuracy / F1	High parameter efficiency ($> 92\%$)	Performance drops under extreme quantization or overly restricted rank (r).
Cheng et al. [6]	2026	LLMs In-Context Learning / SFT	Conversational Emotion Datasets	Macro-F1	Improved multi-turn emotion tracking	Highly sensitive to prompt context; expensive inference latency for broad LLMs.
Liu et al. [10]	2026	Parameter-Efficient Fine-Tuning (LoRA, AdaLoRA, Prefix-Tuning, IA3, BitFit) vs. Full Fine-Tuning (FFT)	GSM8k, ViGGO, SQL, SQuAD 1.1, GLUE/AdvGLUE, Adversarial SQuAD	Accuracy / Macro-F1 / Robustness / Memory Footprint	FFT consistently outperforms PEFT on complex reasoning/SQL and exhibits higher adversarial robustness	PEFT representation capacity is strictly bounded by tuned parameters, shows diminishing marginal gains with larger dataset sizes
Murthy & Kumar [2]	2021	Deep Learning and Transformers (RNN, LSTM, BiLSTM, CNN, BERT)	Dialogues/Conversations (DailyDialog, MELD, EmotionLines, WASSA)	Macro-F1, Jaccard Accuracy, Pearson Correlation (r)	State-of-the-art context and sequential modeling	Computationally expensive; degraded performance on sarcasm, negation, and multi-label edge cases.
Luo et al. [17]	2023	Fine-tuned RoBERTa + 1D CNN-BiLSTM + Residual Blocks	MELD (Text Modality)	Weighted-Average F1	0.6612 (66.12%)	High misclassification in minor classes (disgust, fear, sadness); lacks multi-modal context (audio/visual); sensitive to short, ambiguous utterances.
Jayanthi & Kavitha [16]	2026	TF-IDF + LSTM (Unified Hybrid Framework)	EMO-KNOW (48 emotion classes)	Accuracy / Macro F1-Score	0.954 (95.4%)	Relies on sequential processing of sparse TF-IDF vectors; evaluated primarily on short, informal social media text.
Stigall et al. [7]	2024	Fine-tuned $BERT_{Tiny}$ (Multi-Task Parallel Training)	Aggregated Social Media Datasets (Text)	Accuracy / F1-Score	0.8546 (85.46%)	Class imbalance (underrepresented neutral/worry), data loss from dual-dataloader alignment.

1) Model Selection and Taxonomy: As detailed in Table V, the experimental setup evaluates nine distinct architectures to enable controlled comparative analysis:

This systematic selection isolates key variables across three dimensions:

- **Architectural Family:** Comparing traditional recurrent neural networks (BiLSTM) against transformer-based encoders and decoders under uniform evaluation conditions.
- **Parameter Scale:** Evaluating intra-family scaling behaviors by comparing small versus base/larger variants within encoders (DistilBERT vs. BERT-base), decoders (Qwen2.5-0.5B vs. Qwen2.5-1.5B), and within the Gemma 4 family itself (E2B vs. E4B) [21].
- **Pre-training Objective:** Isolating the impact of training objectives by directly comparing general instruction-tuned Qwen2.5-1.5B against DeepSeek-R1-Distill-Qwen-1.5B, maintaining identical base architecture and parameter count while varying reasoning-oriented fine-tuning.

2) Hardware Environment: Experiments are distributed across local hardware resources representing heterogeneous computational capacities:

- **Standard Workstations:** NVIDIA GTX 1650 (4 GB VRAM) and NVIDIA RTX 2060 (6 GB VRAM) utilized for baseline models, data pre-processing, and lightweight encoder fine-tuning.
- **High-Performance Workstation:** NVIDIA RTX 3080 (10 GB VRAM) allocated for larger encoder runs and parameter-efficient fine-tuning (PEFT/LoRA) of decoder architectures, including the larger Gemma 4 E4B variant (approximately 8B parameters with embeddings).
- **Unified Memory Platform:** Apple MacBook Pro M5 (24 GB Unified Memory) utilized for local execution, low-bit quantized inference testing, and host environment evaluations.

TABLE IV
COMPARATIVE SUMMARY OF PARAMETER-EFFICIENT FINE-TUNING AND AUXILIARY APPROACHES

Reference	Year	Technique	Dataset / Environment	Primary Metric	Reported Result	Identified Limitation
Wang et al. [14]	2025	Comprehensive PEFT Taxonomy Survey (Additive, Reparameterized, Selective, Hybrid, Quantization, Multi-Task)	Multi-Domain (NLP, Vision, Diffusion, MLLM)	Comparative GLUE Benchmarks	LoRA matches full fine-tuning accuracy on RoBERTa-large GLUE tasks while training under 1% of parameters	Lacks principled search strategy for hybrid PEFT combinations; fine-tuned LLMs remain prone to overconfident predictions on small datasets.
Wu et al. [15]	2025	Adversarial Low-Rank Adaptation (ADV-LoRA) with FGSM Perturbations and Self-Supervised Learning	GLUE Subset (SST-2, CoLA, RTE, MRPC) and CNN/DM, XSum	Accuracy / MCC / ROUGE	Outperforms standard LoRA on RoBERTa-large (avg. 85.5 vs. 84.6) and LLaMA-2 7B (CoLA 72.0 vs. 69.4)	Introduces roughly 150% training-time overhead versus standard LoRA; untested on morphologically complex or low-resource languages.
Yang et al. [18]	2023	LLM-Based Contextual Prompt Augmentation for Dataset Auditing	Text-Based Emotion Classification Datasets (Label-Context Alignment)	Human and Empirical Alignment Evaluation	Synthesized context improves alignment between input text and human-annotated emotion labels	Relies on LLM-generated context that may introduce its own biases; does not fine-tune a downstream classifier directly.
Alam et al. [19]	2026	Classical ML, Deep Learning, Transformers (mBERT, XLM-R, Bangla-BERT), and Instruction-Tuned LLMs (GPT-4, Claude, DeepSeek)	BEmoC (Bengali Emotion Corpus, 6 Classes)	Weighted F1-Score	GPT-4 few-shot reaches 82.3% F1, surpassing the best fine-tuned transformer XLM-R (69.7% F1)	LLM inference latency ($\sim$5s/query) hinders deployment; BEmoC's limited size (6,243 instances) constrains generalizability.
Benayas et al. [20]	2025	Fine-Tuned Encoder-Only (RoBERTa) vs. Fine-Tuned/Few-Shot Decoder-Only (LLaMA2-7B via LoRA/QLoRA)	Intent Classification (SNIPS, ATIS, Banking, MASSIVE) and Sentiment Analysis (SST2, Tweet-Eval)	Macro/Micro F1-Score	RoBERTa outperforms LLaMA2 on intent classification (e.g., Banking: 94.27 vs. 70.87 F1); LLaMA2 leads slightly on sentiment analysis (96.21 vs. 91.61 on SST2)	Decoder-only inference is 29x to 108x slower than encoder-only; sensitive to class-name semantics and few-shot example ordering.

TABLE V
TAXONOMY OF EVALUATED MODELS AND EXPERIMENTAL CONTROLS

Model	Architecture Family	Experimental Control
TF-IDF + LogReg	Non-Neural Baseline	Non-deep learning baseline
BiLSTM	Recurrent Neural Net	Recurrent architecture baseline
DistilBERT	Encoder Transformer	Small scale encoder baseline
BERT-base	Encoder Transformer	Scale effect vs. DistilBERT
Qwen2.5-0.5B	Decoder Transformer	Small scale decoder
Qwen2.5-1.5B	Decoder Transformer	Scale effect vs. Qwen2.5-0.5B
Gemma 4 E2B	Decoder Transformer	Family effect vs. Qwen2.5-1.5B
Gemma 4 E4B	Decoder Transformer	Parameter scale effect vs. Gemma 4 E2B
DeepSeek-R1-Distill-Qwen-1.5B	Decoder Transformer	Training objective effect vs. Qwen2.5-1.5B

3) Measurement Protocols: To address severe class imbalance while capturing resource consumption, we evaluate models across dual quality and cost dimensions:

- **Classification Quality:** Macro-F1 serves as the primary metric to ensure underrepresented emotion classes are weighted equally, supplemented by Weighted-F1 and 6×6 confusion matrix error analyses. Simple accuracy is explicitly avoided due to class distribution skew.
- **Computational Cost:** Metrics include total training duration, peak memory footprint (VRAM/RAM), trainable parameter count, and per-sample inference latency.

The central synthesis constructs a Pareto efficiency frontier plotting computational cost (X-axis) against Macro-F1 performance (Y-axis) to identify optimal deployment candidates. This framing is directly motivated by recent evidence in a closely related low-resource emotion classification setting, where an instruction-tuned decoder evaluated in a zero- or few-shot regime achieved the highest reported F1-score but at a substantially higher per-query inference cost than a smaller fine-tuned encoder, illustrating that the highest-quality model and the most cost-efficient model need not coincide [19].

B. Model Selection and Adapter Configuration

To balance model adaptation against hardware constraints across our experimental setups, we apply distinct training strategies based on model architecture and parameter scale.

1) Model Configurations and Training Objectives: We select nine distinct models to evaluate performance across recurrent baselines, encoder architectures, and modern generative decoders:

- **Baseline Models:** We implement a TF-IDF + Logistic Regression model operating on standard n-gram features as a non-neural benchmark. For a recurrent baseline, we train a BiLSTM model using static word embeddings to capture sequential text structures.

- **Encoder Transformers:** We select DistilBERT ($\approx$ 66M parameters) and BERT-Base ($\approx$ 110M parameters) for full supervised fine-tuning (SFT) [4]. Because of their small memory footprints, all parameters across all transformer layers are updated directly using standard cross-entropy loss.
- **Decoder Transformers:** Our decoder models include Qwen2.5-0.5B, Qwen2.5-1.5B, Gemma 4 E2B, Gemma 4 E4B, and DeepSeek-R1-Distill-Qwen-1.5B [21]. To perform fine-tuning within consumer GPU memory limits (such as an NVIDIA RTX 3080 with 10 GB VRAM), the base model weights Θ_0 remain frozen while downstream adaptation is handled through Low-Rank Adaptation (LoRA) [9].

The decision to evaluate encoder and decoder families side by side, rather than treating one as a strict upgrade over the other, follows direct empirical precedent: a controlled comparison between a fine-tuned encoder-only model and a larger decoder-only model on intent classification and sentiment analysis found that the encoder consistently matched or outperformed the decoder at a fraction of the inference latency, with the gap widening as the number of target classes increased [20]. Whether this pattern extends to multi-class emotion recognition once decoder models are adapted via LoRA rather than compared through full fine-tuning or few-shot prompting is one of the open questions this work investigates.

2) LoRA Hyperparameter and Target Module Setup: For decoder models, LoRA adapters are injected into specific linear projection matrices inside the self-attention and feed-forward networks (FFN). Table VI summarizes the key hyper-parameter settings used during training.

TABLE VI
LoRA ADAPTER HYPERPARAMETER CONFIGURATION

Hyperparameter	Value / Configuration
Adapter Rank (r)	16
Scaling Factor (α)	32 ($\alpha/r = 2.0$)
Dropout Rate	0.05
Target Modules (Qwen / Gemma)	Query (W_q), Key (W_k), Value (W_v), Output (W_o)
Target Modules (DeepSeek)	Attention (W_q, W_k, W_v, W_o) + FFN ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$)
Precision	Mixed Precision (FP16 / BF16)
Quantization	4-bit NormalFloat (QLoRA) on low-VRAM GPUs

By setting the adapter rank $r = 16$ and scaling factor $\alpha = 32$, we achieve an effective learning scale while keeping trainable parameters under 1.5% of the base decoder size [9]. In accordance with standard initialization, weight matrix A is initialized using a Gaussian distribution, while matrix B starts at zero to maintain an identity mapping at the start of training.

For our direct comparison between Qwen2.5-1.5B and DeepSeek-R1-Distill-Qwen-1.5B, we keep identical LoRA configurations to isolate the effect of reasoning-focused distillation pre-training against standard instruction tuning.

C. Pre-Processing and Imbalance Mitigation Strategy

1) Text Normalization and Deduplication: Prior to tokenization, the raw corpus is deduplicated to remove exact-duplicate entries that can arise when aggregating community-hosted mirrors of the CARER dataset, preventing the same sample from appearing in both the training and evaluation partitions. Text normalization is kept minimal: standard whitespace and encoding artifacts are cleaned, but punctuation, capitalization, and emoticons are preserved, since these surface-level cues frequently carry affective signal in short, informal social media text [1]. Each model family applies its own native tokenizer (WordPiece for the BERT-based encoders, byte-pair encoding for the Qwen, Gemma, and DeepSeek decoders) directly on this normalized text, without a shared vocabulary or intermediate representation across families.

2) Class-Weighted Loss Under Frozen Backbones: As previously noted, standard cross-entropy loss favors majority classes and can mask poor recall on underrepresented emotions [1], and this imbalance is only partially addressed by existing PEFT frameworks, since the base decoder weights remain frozen throughout adaptation [9]. Because a frozen backbone does not prevent reweighting the loss computed at the classification head, this work applies inverse-frequency class weighting to the cross-entropy objective uniformly across all eight models, whether the head sits atop a fully fine-tuned encoder or a LoRA-adapted decoder. This keeps the imbalance-mitigation strategy constant across the three axes under study, so that any observed differences in Macro-F1 can be attributed to architectural family, parameter scale, or training objective rather than to inconsistent handling of class imbalance between models.

D. Evaluation Protocol and Statistical Validation

1) Data Partitioning: The corpus is split once into training, validation, and test partitions ($80\%/10\%/10\%$), stratified by emotion label to preserve the original class proportions described in Section IV.A across all three partitions. This fixed partition is shared across all eight models to avoid the risk, specific to text classification tasks, of near-duplicate samples leaking between training and evaluation splits when partitions are regenerated independently per model; the validation split is used exclusively for early stopping and hyperparameter selection, never for reporting final results.

2) Variance Across Random Seeds: Because parameter-efficient fine-tuning has been shown to introduce measurable instability relative to full fine-tuning, particularly under the sparse trainable-parameter regime that LoRA and QLoRA rely on [15], each decoder configuration is trained across three random seeds, with the held-out test set evaluated independently for each run. Results are reported as mean $\pm$ standard deviation on Macro-F1 rather than as a single-run point estimate, so that the reported quality-cost trade-off reflects the typical behavior of each configuration rather than a favorable outlier.

3) Statistical Comparison Between Models: Pairwise comparisons that isolate a single axis, such as DistilBERT against BERT-Base for parameter scale or Qwen2.5-1.5B against DeepSeek-R1-Distill-Qwen-1.5B for training objective, are evaluated with a paired significance test on a per-sample basis, following the same paired-testing approach used to validate PEFT robustness claims in the reviewed literature [15]. Given the small number of classes and runs involved, results are cross-checked with a non-parametric alternative when the normality of the underlying score distribution cannot be assumed, in line with the Kruskal-Wallis and post-hoc testing procedure used to evaluate LLM-based emotion classification differences in related work [18]. A difference between two configurations is treated as informative only when it clears this statistical threshold, rather than being read directly off the raw Macro-F1 gap.

VI. Discussion & Open Research Directions

VII. Conclusion

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